Predictive Model Construction and Risk Assessment for Acute Coronary Syndrome Patients with Hypertension

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Submitted to Journal:
Frontiers in Cardiovascular Medicine

Specialty Section:
Coronary Artery Disease

Article type:
Original Research Article

Manuscript ID:
1541098

Received on:
12 Dec 2024

Journal website link:
www.frontiersin.org

Scope Statement

This study aims to develop a nomogram model to predict the risk in ACS patients with hypertension, providing clinicians with a tool for early diagnosis, personalized treatment, and prognostic evaluation. Data were collected from ACS patients at Huangshi Aikang Hospital between 2018 and 2023. Statistical analyses were performed using SPSS (27.0.1) and R software (4.3.2), with statistical significance set at P < 0.05. A total of 980 ACS patients were included in the study. Among the three clinical subtypes, 592 patients (60.4%) had UA, which was the most prevalent. The hypertensive group comprised 682 ACS patients (69.59%), with a mean age of 64.93 ± 9.51 years. Significant differences between hypertensive and non-hypertensive groups were found in sex (P=0.001), age (P<0.001), clinical subtype (P<0.001), and several clinical and laboratory parameters, including creatinine (Cr) (P<0.001), left ventricular ejection fraction (LVEF) (P=0.049), left ventricular posterior wall thickness (LVPW) (P=0.003), CK-MB (P=0.019), AST (P=0.028), total cholesterol (TC) (P=0.035), LDL-C (P=0.007), and APOB (P=0.005). Using LASSO regression, nine variables were selected for multivariate logistic regression analysis, leading to the construction of the nomogram model. This model can assist clinicians in early diagnosis, personalized treatment, and prognostic evaluation.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

Credit Author Statement

Haozhen Bai: Data curation, Investigation, Methodology, Software, Validation, Writing – original draft. Changwu Feng: Conceptualization, Data curation, Investigation, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft. Li Song: Data curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. Jumin Xie: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. Zixuan Yang: Data curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft.

Keywords

Acute Coronary Syndromes, Hypertension, LASSO, Nomogram model, Risk Assessment

Abstract

Word count: 335

Cardiovascular disease (CVD) remains the leading cause of death worldwide, according to global statistics from the WHO and GBD, with the incidence of acute coronary syndromes (ACS) continuing to rise annually. This study aims to develop a nomogram model to predict the risk in ACS patients with hypertension, providing clinicians with a tool for early diagnosis, personalized treatment, and prognostic evaluation.Data were collected from ACS patients at Huangshi Aikang Hospital between 2018 and 2023. Patient characteristics, including age, gender, hypertension history, initial blood test results, and cardiac Doppler ultrasound findings, were recorded. ACS diagnosis followed the 2019 revised Guidelines for the Diagnosis and Treatment of Acute ST-Segment Elevation Myocardial Infarction (STEMI) by the Chinese Society of Cardiology. The 2024 Revised Guidelines for the Diagnosis and Treatment of Non-ST-Segment Elevation Acute Coronary Syndromes from the Chinese Journal of Cardiovascular Diseases were used for NSTEMI and unstable angina (UA) diagnoses. Statistical analyses were performed using SPSS (version 27.0.1) and R software (version 4.3.2), with statistical significance set at P < 0.05.A total of 980 ACS patients were included in the study. Among the three clinical subtypes, 592 patients (60.4%) had UA, which was the most prevalent. The hypertensive group comprised 682 ACS patients (69.59%), with a mean age of 64.93 ± 9.51 years. Significant differences between hypertensive and non-hypertensive groups were found in sex (P=0.001), age (P<0.001), clinical subtype (P<0.001), and several clinical and laboratory parameters, including creatinine (Cr) (P<0.001), left ventricular ejection fraction (LVEF) (P=0.049), left ventricular posterior wall thickness (LVPW) (P=0.003), CK-MB (P=0.019), AST (P=0.028), total cholesterol (TC) (P=0.035), LDL-C (P=0.007), and APOB (P=0.005). Using LASSO regression, nine variables were selected for multivariate logistic regression analysis, leading to the construction of the nomogram model. The calibration curve, Hosmer-Lemeshow test, ROC curve, decision curve, and clinical impact curve all demonstrated the model's high quality. Conclusions A high-quality predictive nomogram model for assessing the risk of ACS in patients with hypertension has been developed. This model can assist clinicians in early diagnosis, personalized treatment, and prognostic evaluation.

Funding information

This study was supported by Key Project in the Scientific Research Plan of the Hubei Province Education Department (D20234505) and Open fund research project of Huangshi Key Laboratory of Molecular Diagnosis and Individualized Therapy (FZ2024005).

Funding statement

The author(s) declare that financial support was received for the research, authorship, and/or publication of this article.

Ethics statements

Studies involving animal subjects

Generated Statement: No animal studies are presented in this manuscript.

Studies involving human subjects

Generated Statement: The studies involving humans were approved by Bioethical Safety Committee of Hubei Polytechnic University (with the license number: yxy-20240101). The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation was not required from the participants or the participants' legal guardians/next of kin in accordance with the national legislation and institutional requirements.

Inclusion of identifiable human data

Generated Statement: No potentially identifiable images or data are presented in this study.

Data availability statement

Generated Statement: The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author/s.

Generative AI disclosure

No Generative AI was used in the preparation of this manuscript.

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Abstract:

Background

Cardiovascular disease (CVD) remains the leading cause of death worldwide, according to global statistics from the WHO and GBD, with the incidence of acute coronary syndromes (ACS) continuing to rise annually. This study aims to develop a nomogram model to predict the risk in ACS patients with hypertension, providing clinicians with a tool for early diagnosis, personalized treatment, and prognostic evaluation.

Methods

Data were collected from ACS patients at Huangshi Aikang Hospital between 2018 and

2023. Patient characteristics, including age, gender, hypertension history, initial blood test results, and cardiac Doppler ultrasound findings, were recorded. ACS diagnosis followed the 2019 revised Guidelines for the Diagnosis and Treatment of Acute ST-Segment Elevation Myocardial Infarction (STEMI) by the Chinese Society of Cardiology. The 2024 Revised Guidelines for the Diagnosis and Treatment of Non-ST-Segment Elevation Acute Coronary Syndromes from the Chinese Journal of Cardiovascular Diseases were used for NSTEMI and unstable angina (UA) diagnoses. Statistical analyses were performed using SPSS (version 27.0.1) and R software (version 4.3.2), with statistical significance set at $P < 0.05$.

Results

A total of 980 ACS patients were included in the study. Among the three clinical subtypes, 592 patients (60.4%) had UA, which was the most prevalent. The hypertensive group comprised 682 ACS patients (69.59%), with a mean age of 64.93 ± 9.51 years. Significant differences between hypertensive and non-hypertensive groups were found in sex ($P=0.001$), age ($P<0.001$), clinical subtype ($P<0.001$), and several clinical and laboratory parameters, including creatinine (Cr) ($P<0.001$), left ventricular ejection fraction (LVEF) ($P=0.049$), left ventricular posterior wall thickness (LVPW) ($P=0.003$), CK-MB ($P=0.019$), AST ($P=0.028$), total cholesterol (TC) ($P=0.035$), LDL-C ($P=0.007$), and APOB ($P=0.005$). Using LASSO regression, nine variables were selected for multivariate logistic regression analysis, leading to the construction of the nomogram model. The calibration curve, Hosmer-Lemeshow test, ROC curve, decision curve, and clinical impact curve all demonstrated the model's high quality.

Conclusions

A high-quality predictive nomogram model for assessing the risk of ACS in patients with hypertension has been developed. This model can assist clinicians in early diagnosis, personalized treatment, and prognostic evaluation.

Keywords: Acute coronary syndromes, Hypertension, LASSO, Nomogram model,

1. Introduction

According to global statistics released by the World Health Organization (WHO) in 2021, cardiovascular disease (CVD) is the leading cause of death worldwide, with approximately 17.9 million deaths in 2019, accounting for 32% of all fatalities. More than three-quarters of CVD-related deaths occur in low- and middle-income countries [1]. CVD is the primary cause of death in China, with estimated CVD-related deaths rising from 3.09 million in 2005 to 4.58 million in 2020, accounting for 40% of total deaths [2, 3].

Acute coronary syndromes (ACS) are severe clinical manifestations of CVD, typically caused by acute blockage or reduced blood flow in the coronary arteries [4]. ACS encompasses unstable angina (UA), non–ST-segment elevation myocardial infarction (NSTEMI) and ST-segment elevation myocardial infarction (STEMI) [5].

UA is characterized by intravascular thrombosis without complete obstruction, leading to temporary myocardial ischemia and severe symptoms without myocardial necrosis [6]. NSTEMI results from partial coronary artery obstruction, causing myocardial damage without fully blocking blood flow [7] . STEMI is the most severe form of ACS, involving complete coronary artery blockage and extensive myocardial necrosis [8]. Acute myocardial infarction typically arises from either complete or partial coronary artery blockage, leading to myocardial ischemia and necrosis [9, 10].

Common causes of ACS include atherosclerotic plaque rupture, thrombosis, coronary artery spasm, hypertension, and stressors such as infections, surgeries, or excessive physical exertion [11-13]. Additional risk factors include drug abuse (e.g., cocaine, amphetamines), unhealthy lifestyle habits (e.g., smoking, excessive alcohol consumption, high-fat diets, physical inactivity), and underlying conditions like diabetes, hyperlipidemia, and sleep apnea syndrome [4, 12]. Effective risk reduction strategies include adopting a healthy diet, quitting smoking, controlling hypertension and diabetes, and engaging in regular physical activity, which can significantly lower the risk of ACS [14, 15]. ACS result from the interplay of multiple complex factors, including unpredictable causes, which complicate risk assessment. ACS is a significant contributor to global morbidity and mortality, and its treatment, along with associated costs, poses an increasing burden on society [16].

The clinical symptoms of acute coronary syndrome (ACS) are diverse and primarily associated with myocardial ischemia and infarction [4]. These symptoms often vary depending on the severity of the condition and individual patient differences. Common symptoms include chest pain or discomfort, often accompanied by nausea, vomiting, fatigue, cold sweats, and shortness of breath [17-19]. Resting chest discomfort is the most common initial symptom of ACS, occurring in approximately 79% of men and 74% of women. However, about 40% of men and 48% of women present with non-specific symptoms, such as isolated or combined dyspnea with chest pain. The pain or discomfort associated with ACS can occur during physical exertion or at rest and is typically diffuse rather than localized [13, 20]. Elderly individuals, women, and patients with diabetes may present with atypical symptoms, such as dizziness, fainting, unexplained fatigue or weakness, upper abdominal pain, nausea, vomiting, belching, or pain in the back, shoulders, neck, or jaw [17, 18]. Additional clinical manifestations of ACS include hypertension or shock, arrhythmias, acute left heart failure, and silent myocardial ischemia [21-23].

More than half of patients with ACS have a history of hypertension, with a higher prevalence observed in women compared to men [24-26]. The incidence of hypertension among ACS patients is steadily increasing [27, 28]. If left uncontrolled, hypertension can severely impair the effectiveness of rehabilitation treatments.

Constructing a clinical prediction model for ACS enables systematic and scientific analysis, facilitating more accurate estimation of ACS incidence [29, 30]. Notably, the Least Absolute Shrinkage and Selection Operator (LASSO) algorithm is employed to establish a regression model [31]. This method incorporates a penalty function to effectively identify variables with significant predictive impact, thereby enhancing the model's accuracy and reliability. A nomogram will then be used to visualize the analysis results, transforming complex, abstract data into an intuitive graphical format that simplifies interpretation and provides clearer predictive insights [32].

Given that ACS is influenced by multiple factors, single-factor predictions are insufficient for accurate risk assessment. Thus, predictive models must incorporate multiple relevant factors to comprehensively evaluate the risk of adverse outcomes in ACS patients. Currently, biomarkers and risk prediction models for hypertension in patients with ACS are scarce.

This study aimed to analyze the general clinical data, physiological and biochemical indicators, and cardiac color Doppler ultrasound results of ACS patients. Using LASSO regression analysis, key risk factors for hypertension in ACS patients were identified. Based on these findings, a predictive nomogram model was constructed and evaluated to assess its clinical utility. This study seeks to enhance the early prediction accuracy for hypertension in ACS patients, enable the development of personalized treatment plans, and provide an effective reference for prognostic analysis.

2. Case information and methods

2.1 Case information

2.1.1 Cases collection

A total of 1,526 ACS patients who underwent percutaneous coronary intervention therapy at Huangshi Aikang Hospital between December 1, 2018, and November 30, 2023, were enrolled in the study. After excluding 546 cases with incomplete information, 980 valid cases were included for this analysis.

2.1.2 Screening criteria

Inclusion Criteria:

(1) Patients diagnosed with ACS, with complete laboratory test results;

(2) Presence of at least two of the following three symptoms: ischemic chest pain, chest tightness, dynamic electrocardiogram changes, and elevated myocardial enzyme levels;

(3) Adequate cognitive function;

(4) Voluntary participation in the study.

Exclusion Criteria:

(1) Contraindication to PCI;

(2) Chest pain or chest tightness due to non-coronary conditions (e.g., valvular heart disease, aortic dissection, acute pulmonary embolism, pericarditis, pneumonia,

pneumothorax, pleurisy) or gastrointestinal disorders (e.g., reflux esophagitis);

(3) Severe liver, kidney, or organ failure, or malignancy;

(4) Presence of psychiatric disorders;

(5) Incomplete clinical trial data.

2.2 Contents and methods

2.2.1 Data collection

General information, including age, gender, smoking history, and history of hypertension, was recorded. Blood test indicators were also collected upon admission, including white blood cell count (WBC), red blood cell count (RBC), plasma prothrombin time (PT), plasma fibrinogen (FIB), serum alanine aminotransferase (ALT), serum aspartate aminotransferase (AST), creatinine (Cr), triglycerides (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein A-I (ApoA-I), apolipoprotein B (ApoB), fasting blood glucose (GLU), blood uric acid (BUA), and creatine kinase-MB (CK-MB). Additionally, the results of color Doppler ultrasonography were collected, including left ventricular ejection fraction (LVEF) and left ventricular posterior wall thickness (LVPW).

2.2.2 Judgement standard

(1) Diagnostic criteria for ACS

Diagnostic criteria for STEMI: The 2019 revised Guidelines for the Diagnosis and Treatment of Acute ST-Segment Elevation Myocardial Infarction from the Chinese Society of Cardiology.

Diagnostic criteria for NSTEMI and UA: The 2024 Revised Guidelines for the Diagnosis and Treatment of Non-ST-Segment Elevation Acute Coronary Syndromes published in the Chinese Journal of Cardiovascular Diseases.

(2) Diagnostic criteria for each risk factor

Diagnostic Criteria for Hypertension: According to the 2020 Global Hypertension Practice Guidelines issued by the International Society of Hypertension (ISH), hypertension is defined as a systolic blood pressure ≥140 mmHg and/or diastolic blood pressure ≥90 mmHg, measured multiple times on different days, without the use of antihypertensive medications. (1 mmHg = 0.133 kPa). This diagnosis applies to patients with a confirmed history of hypertension who are receiving antihypertensive treatment.

Age Groups: Based on the age classification for hypertension in the elderly, as outlined in practical internal medicine textbooks, age groups were categorized as follows: under 49 years, 50–64 years, 65–79 years, and 80 years or older.

2.3 Statistical analysis

SPSS (version 27.0.1) and R software (version 4.3.2) were used for statistical analysis. The Kolmogorov-Smirnov (K-S) test was applied to assess the distribution characteristics of measurement data. Measurement data with a normal distribution were expressed as mean ± standard deviation (x ± s) and compared between groups using the independent sample *t*-test. For data with a non-normal distribution, the median (interquartile range) was used, and group comparisons were conducted using the Mann-Whitney U test.

Categorical data were expressed as counts (n) or percentages (%) and analyzed using the chi-square test. Statistically significant variables identified through univariate analysis were subjected to LASSO regression analysis using the "glmnet" R package to select the optimal predictive variables while minimizing the risk of model overfitting.

Multivariate logistic regression analysis was performed on the selected variables using the "rms" R package, and a nomogram prediction model was developed to estimate the risk of ACS combined with hypertension. The calibration ability of the model was assessed using the Hosmer-Lemeshow goodness-of-fit test to ensure its clinical applicability.

Receiver operating characteristic (ROC) curves were generated, and the area under the curve (AUC) was calculated to evaluate the model's discriminative performance. Decision curve analysis (DCA) and clinical impact curves (CIC) were used to further assess the model’s predictive accuracy and clinical utility. A level of $P < 0.05$ was considered statistically significant.

3 Results

3.1 Analysis of Clinical Characteristics of ACS Cases at Aikang Hospital, Huangshi (2018–2023)

This study included 980 patients with ACS who met the screening criteria. The annual distribution of ACS cases from 2018 to 2023 is presented in (Figure 1A). Among the three clinical subtypes of ACS, the number of UA cases was the highest, accounting for 60.4% (N=592), and exhibited an increasing trend over the years (Figure 1B). Based on the presence or absence of hypertension, the 980 ACS patients were categorized into a non-hypertensive group and a hypertensive group. The non-hypertensive group comprised 298 patients with an average age of 62.37 ± 10.41 years, including 77.52% males (N=231) and 22.48% females (N=67) (Figure 1C). The hypertensive group included 682 patients with a mean age of 64.93 ± 9.51 years, of whom 67.45% were males (N=460) and 32.55% were females (N=222) (Figure 1D). From 2018 to 2023, the total number of ACS cases increased annually, with males accounting for a larger proportion (N=691, 70.51%). ACS Patients aged 65–79 with hypertension (N=375, 54.99%) demonstrated the highest risk (Table 1).

3.2 Risk Factors Analysis in ACS Patients with or without Hypertension

Significant differences were observed in sex (P=0.001), age (P<0.001), and clinical subtypes (P<0.001) of ACS patients between the non-hypertensive and hypertensive groups (Table 1). Additionally, Cr (P<0.001), LVEF (P=0.049), and LVPW (P=0.003) levels were significantly higher in the hypertensive group compared to the non-hypertensive group, demonstrating notable statistical differences. Conversely, CK-MB (P=0.019), AST (P=0.028), TC (P=0.035), LDL-C (P=0.007), and APOB (P=0.005) levels were significantly lower in the hypertensive group than in the non-hypertensive group, with statistically significant differences (Table 2). However, no significant differences were found in WBC, RBC, PT, FIB, ALT, BUA, TG, or HDL-C levels between the two groups (Table 2).

3.3 LASSO regression analysis

The 11 indicators with *P*<0.05 in the univariate analysis were selected as independent variables, while the presence of hypertension was set as the dependent variable (Tables 1&2). LASSO regression analysis was then performed. To identify the optimal penalty coefficient λ, enhance model performance, and minimize the influence of irrelevant factors, 10-fold cross-validation was conducted. The optimal λ value was determined as the point (vertical dotted line on the right) corresponding to one standard deviation above the minimum mean square error (Figure 2A&B). Based on λ value, nine predictors were identified: Sex, Age, CK-MB, Cr, LDL-C, APOB, LVEF, LVPW, and ACS clinical subtypes.

3.4 Construction and evaluation of nomogram model

The nine selected predictors were included in a multivariate logistic regression analysis to construct a nomogram model (Figure 2C). The bootstrap method, with 1,000 resampling iterations, was used for internal validation of the model. The calibration curve closely aligned with the ideal curve, with an absolute error of 0.023, indicating good predictive accuracy of the nomogram model (Figure 3A). The Hosmer-Lemeshow test yielded P = 0.4024 demonstrating a good model fit (P>0.05).

The ROC curve analysis further evaluated the model's efficacy in predicting the risk of ACS combined with hypertension, resulting in an AUC of 0.6506 (95% CI: 0.6113 to 0.6879), indicating the model's satisfactory performance (Figure 3B). Decision curve analysis (DCA) indicated that when the model's threshold probability ranged between 5% ‑ 47% and exceeded 55%, the overall net benefit was superior to intervening in all or none of the cases, highlighting its strong clinical utility (Figure 3C). Additionally, the clinical impact curve (CIC) demonstrated that when the model's threshold value exceeded 0.4, its predictions for hypertension in patients with acute coronary syndrome closely matched the actual population with these conditions (Figure 3D). Overall, multiple validation methods confirmed that the nomogram model developed in this study effectively predicts and assesses clinical risk in patients with acute coronary syndrome and hypertension.

4 Discussion

According to the WHO and the Global Burden of Disease (GBD) Study, cardiovascular diseases was the leading cause of deaths globally each year [1]. Among these, ACS-related conditions, such as myocardial infarction and UA, are the leading contributors, causing an estimated 7.5 million deaths annually [33, 34]. Developing a reliable predictive model based on the risk factors for ACS could serve as a valuable tool for guiding early clinical diagnosis and treatment.

This study analyzed clinical data from 980 patients with ACS treated at Aikang Hospital in Huangshi between 2018 and 2023. Key variables included age, gender, history of hypertension, initial blood test results upon admission, and cardiac color Doppler ultrasound findings. Using these data, a nomogram model based on LASSO regression analysis was developed to offer a novel approach for risk assessment, early diagnosis, and treatment of patients with acute coronary syndrome and hypertension.

Among the 980 ACS patients, 592 presented with UA, highlighting its high prevalence in this population and underscoring the need for heightened clinical attention. Early diagnosis and intervention are critical, as UA represents a precursor stage of ACS. Without timely and effective treatment, it can rapidly progress to myocardial infarction or other severe complications. Given the high incidence of UA in ACS, appropriate acute-phase management is essential. This includes antiplatelet and anticoagulant therapies, alongside pharmacological interventions such as nitrates and beta-blockers, to mitigate ischemic risk and prevent disease progression [35, 36]. While UA often presents with severe symptoms, not all cases lead to myocardial infarction. Clinicians must perform thorough risk assessments and implement early interventions using tools such as electrocardiography (ECG), blood biomarkers, and coronary angiography [37].

In this study, 682 patients with ACS were found to have hypertension, accounting for 69.59% of all cases, indicating that hypertension is a prevalent and serious comorbidity. Hypertension is recognized as a significant risk factor for ACS, as it increases cardiac workload, accelerates atherosclerosis, damages vascular endothelium, and triggers inflammatory responses, thereby creating a pro-thrombotic environment and elevating the risk of coronary events [38, 39]. Moreover, hypertensive patients often exhibit additional cardiovascular risk factors, such as hyperglycemia and hyperlipidemia, which exacerbate cardiac strain and vascular damage [40].

Hypertension not only raises the likelihood of developing ACS but also adversely impacts the prognosis of ACS patients. For those with a history of hypertension, strict blood pressure control is crucial to reducing the incidence of subsequent cardiovascular events. Given the heterogeneity of ACS patients, antihypertensive treatment should be individualized. Commonly used medications include ACE inhibitors, ARBs, beta-blockers, and calcium channel blockers, which not only help control blood pressure but also improve coronary perfusion and reduce the risk of adverse cardiac events [41, 42].

In addition to pharmacological management, lifestyle modifications play a critical role in improving outcomes for ACS patients with hypertension. These include adopting a healthy diet, engaging in regular physical activity, achieving weight reduction, and minimizing salt intake [43, 44]. Such measures are essential for blood pressure control and improving long-term prognosis.

In this study, serum biomarkers, including CK-MB, Cr, LDL-C, and ApoB, were utilized to construct a risk assessment model for ACS. CK-MB, a highly sensitive marker of myocardial injury, typically begins to rise 2–4 hours after myocardial infarction, peaks within 24 hours, and gradually normalizes. It is invaluable for diagnosing acute myocardial infarction when combined with clinical symptoms and other cardiac markers (e.g., troponin) [45-47].

Cr serves as a key indicator of renal function. ACS patients frequently experience acute kidney injury, particularly after interventions such as percutaneous coronary intervention (PCI) [48, 49]. Elevated Cr levels suggest impaired renal function, which is more common in elderly patients or those with chronic kidney disease. Renal insufficiency can influence clinical decision-making and is strongly associated with poor ACS outcomes, including increased mortality, heart failure, and recurrent cardiovascular events [50, 51].

LDL-C is a major risk factor for atherosclerosis and a key contributor to coronary heart disease (CHD) and ACS [52]. Elevated LDL-C levels are commonly observed in ACS patients. Long-term control of LDL-C to target levels (<70 mg/dL) is critical for reducing cardiovascular risk [53]. Statins are commonly used to lower LDL-C, slow atherosclerosis progression, and prevent thrombosis, making LDL-C management a cornerstone of ACS treatment to prevent recurrence [54, 55].

ApoB is an early risk assessment marker for ACS, providing a more accurate reflection of atherogenic particle number than LDL-C [56]. High ApoB levels are closely linked to increased cardiovascular risk [57]. Controlling ApoB levels can mitigate atherosclerosis progression and reduce the incidence of cardiovascular events [58, 59]. Serum biomarkers play a pivotal role in diagnosing ACS, guiding treatment strategies, and assessing prognosis, highlighting their significance in comprehensive cardiovascular care.

In ACS patients, Left Ventricular Ejection Fraction (LVEF) is a critical measure for evaluating myocardial damage and cardiac function [60]. A reduced LVEF (<40%) indicates significant myocardial injury and an increased risk of heart failure [61, 62]. Patients with ACS and low LVEF require vigilant monitoring for heart failure symptoms and early interventions, such as treatment with ACE inhibitors, beta-blockers, or angiotensin receptor-neprilysin inhibitors (ARNIs), to improve cardiac function [63].

Left ventricular posterior wall thickness (LVPW) is commonly used to assess cardiac structure. LVPW thickening is frequently observed in conditions such as chronic hypertension, cardiac remodeling following myocardial infarction, and cardiomyopathy [64, 65]. In ACS patients, significant LVPW thickening suggests increased cardiac pressure and potential left ventricular diastolic dysfunction. This thickening is associated with an elevated risk of heart failure, underscoring its predictive value in risk stratification for ACS patients [66, 67]. Together, LVEF and LVPW provide essential insights into cardiac function, the extent of myocardial damage, and the overall severity of ACS, guiding clinical decision-making and management strategies.

In this study, we utilized LASSO regression analysis for the first time to identify risk factors for ACS complicated by hypertension. We then constructed a nomogram model based on multifactor logistic regression, providing an intuitive and effective tool for early diagnosis, personalized treatment, and prognosis assessment of ACS patients with hypertension. This model demonstrates significant clinical value, particularly the internally validated version, which offers a scientific basis for clinicians in making treatment decisions. It aids in the early identification of high-risk patients with ACS combined with hypertension, facilitating timely personalized treatment, improving patient outcomes, and extending survival times.

Nonetheless, this study has four key limitations: (1) The data source is singular, the sample size is limited, and there is potential regional bias, which diminishes the generalizability of the risk prediction model. (2) The absence of long-term follow-up data for ACS patients constrains the model's ability to assess the prognosis of hypertensive patients with ACS. (3) Control of potential confounding factors needs reinforcement, and the accuracy of current technical methods in data analysis is limited. (4) There is a lack of external validation datasets to further evaluate the model's predictive performance.

Future research will focus on expanding the sample size, conducting multi-center and multi-regional data collection and analysis, undertaking long-term prognostic follow-up of ACS patients with hypertension, and implementing more stringent data quality controls as well as comprehensive model validation, to enhance the stability and reliability of the model.

5 Conclusion

Based on the primary data of 980 patients with ACS treated at Aikang Hospital in Huangshi from 2018 to 2023, this study developed a nomogram model for the risk assessment of ACS combined with hypertension. The aim is to offer references and guidance for the early diagnosis, personalized treatment, and prognostic evaluation of ACS in conjunction with hypertension.

Abbreviation

Abbreviation	Full name
ACS	Acute coronary syndrome
ALT	Alanine Aminotransferase
APOB	Apolipoprotein B
AST	Aspartate transaminase
AUC	Area under the curve
BUA	Blood uric acid
CIC	Clinical impact curve
CK-MB	Creatine kinase-MB
Cr	Creatinine
DCA	Decision curve analysis
FGB	Fasting blood glucose
FIB	Fibrinogen
HDL-C	High-density lipoprotein cholesterol
LASSO	Least absolute shrinkage and selection operator
LDL-C	Low-density lipoprotein cholesterol
LVEF	Left ventricular ejection fractions
LVPW	Left ventricular posterior wall thickness
NSTEMI	Non-ST-elevation myocardial infarction
OR	Odd ratio
PT	Prothrombin time
RBC	Red blood cell
ROC	Receiver operating characteristic
STEMI	ST-elevation myocardial infarction
TC	Total cholesterol
TG	Tri-glyceride
UA	Unstable angina
WBC	White blood cell

Acknowledgements

Thanks to all the participants in this study for their kind support and coordination. Special thanks to Wenting Xia from Huangshi Aikang Hospital for her efforts in data collection.

Patient consent statement

All participants in this study provided written informed consent. The participants in this study who had limited educational background were adequately informed and provided written informed consent through their designated surrogate decision-maker or legally authorized representative (LAR).

Disclosure

The authors have nothing to disclosure.

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Authors' contributions

J. X and C. F conceived and designed the study. W. X collected and provided the data. J. X, L. S, Z. Y and H. B investigated and analyzed the data. J. X and C. F supervised the study. J. X, L. S, and Z. Y wrote the draft. J. X wrote and revised the manuscript. J. X, L. S, Z. Y, H. B, and C. F read and agreed to publish the paper.

Funding

This study was supported by Key Project in the Scientific Research Plan of the Hubei Province Education Department (D20234505) and Open fund research project of Huangshi Key Laboratory of Molecular Diagnosis and Individualized Therapy (FZ2024005).

Data availability and materials

The processed data was available in the paper, and raw data is freely serviced from first and corresponding author.

Ethics Declarations

Ethics approval and consent to participate

All procedures involving human subjects/patients were approved by the Bioethical Safety Committee of Hubei Polytechnic University (with the license number: yxy-20240101). The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. All patients in this study had been informed, and they consented to sharing the data.

Consent for publication

Not applicable

Competing interests

The authors declare no conflict of interest.

Reference

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Table 1: Primary information of 980 ACS patients.

Factors	No.of cases		Non-hypertension (n=298)	Hypertension (n=682)	x^2	*P*
Sex					10.11	0.001
	Male	691	231(77.52%)	460(67.45%)		
	Female	289	67(22.48%)	222(32.55%)		
Age (year)					28.404	<0.001
	<50	82	40(13.42%)	42(6.16%)		
	50-64	378	134(44.97%)	244(35.78%)		
	65-79	491	116(38.93)%	375(54.99%)		
	≥80	29	8(2.68%)	21(3.08%)		
ACS Subtypes					14.546	<0.001
	UA	592	164(55.03%)	428(62.76%)		
	STEMI	180	76(25.50%)	104(15.25%)		
	NSTEMI	208	58(19.46%)	150(21.99%)		

Table 2: Blood test and color Doppler ultrasonography results of 980 ACS patients.

Factors	Non-hypertension (n=298)	Hypertension (n=682)	P
LVEF (%)	58.13±8.75	59.30±8.09	0.049
AST (U/L)	62.61±93.35	49.13±74.62	0.028
TC(mmol/L)	4.23±1.23	4.05±1.11	0.02
Cr(μmol/L)	81.17±23.63	96.16±108.60	<0.001
LVPW (mm)	9.21±0.95	9.40±0.87	0.003
LDL-C(mmol/L)	2.30±0.94	2.14±083	0.007
APOB(g/L)	0.86±0.29	0.81±0.25	0.005
CKMB(ng/mL)	28.02±57.80	19.04±48.39	0.019
PT(s)	11.75±1.29	11.84±1.46	0.407
FIB(g/L)	3.67±0.85	13.37±254.50	0.511
ALT (U/L)	29.12±25.84	28.88±38.70	0.924
TG(mmol/L)	1.59±1.137	1.74±1.39	0.095
BUA(mmol/L)	492.83±2186.55	370.24±106.49	0.334
GLU(mmol/L)	6.93±3.20	6.76±2.74	0.435
HDL-C(mmol/L)	1.23±0.31	1.20±0.30	0.203
WBC(×10^9/L)	8.1±3.181	7.75±3.085	0.109
RBC(×10^12/L)	4.43±0.53	4.41±0.55	0.687

Figure legend

Figure 1. An overview of the primary characteristics of ACS patients. (A) Annual distribution of ACS patients treated at Huangshi Aikang Hospital from 2018 to 2023. (B) Annual distribution of the number of cases across the three clinical subtypes of ACS from 2018 to 2023. (C&D) Age distribution, gender proportion, and annual distribution of ACS patients divided into non-hypertensive and hypertensive groups from 2018 to 2023.

Figure 2. Risk factors screening and nomogram model construction in ACS patients with hypertension. (A) LASSO regression path plot for variable selection. The X-axis represents the log(λ) of the penalty coefficient, while the Y-axis shows the binomial deviation. The dashed vertical line at the right indicates λ-se, which corresponds to one standard error above the minimum λ. At λ = 0.02170939, the model achieves optimal fit with a reduced number of predictor variables. (B) Cross-validation plot for LASSO regression variables. Each curve illustrates the coefficient change trajectory of an independent variable, with the Y-axis representing the coefficient value. The lower X-axis shows log(λ), while the upper X-axis indicates the number of non-zero coefficients in the model at each point. (C) Nomogram model for risk assessment of ACS patients with hypertension. Each patient index is assigned a score based on the top line segment. The scores for all indicators are summed to yield a total score at the bottom line, which corresponds to assess the risk in ACS patients with hypertension.

Figure 3. Illustration of the calibration curve, ROC curve, decision curve analysis (DCA), and clinical impact curve (CIC) for the nomogram model of ACS combined with hypertension. (A) The calibration curve of the nomogram demonstrates the alignment between predicted and actual probabilities. The Hosmer-Lemeshow test yielded a p-value of 0.4024 ($P>0.05$), indicating a good fit for the model. (B) The ROC curve of the nomogram model shows an Area Under the Curve (AUC) of 0.651, suggesting moderate model performance. (C) The decision curve analysis of the nomogram model demonstrates significant clinical benefit in accurately distinguishing between patients with ACS and those with ACS combined with hypertension, within a reasonable risk threshold probability range. (D) The clinical impact curve of the nomogram model displays the clinical effects at various probability thresholds. As the threshold varies, the model's agreement between predicted and actual cases of ACS with hypertension is evaluated.

Figure 1

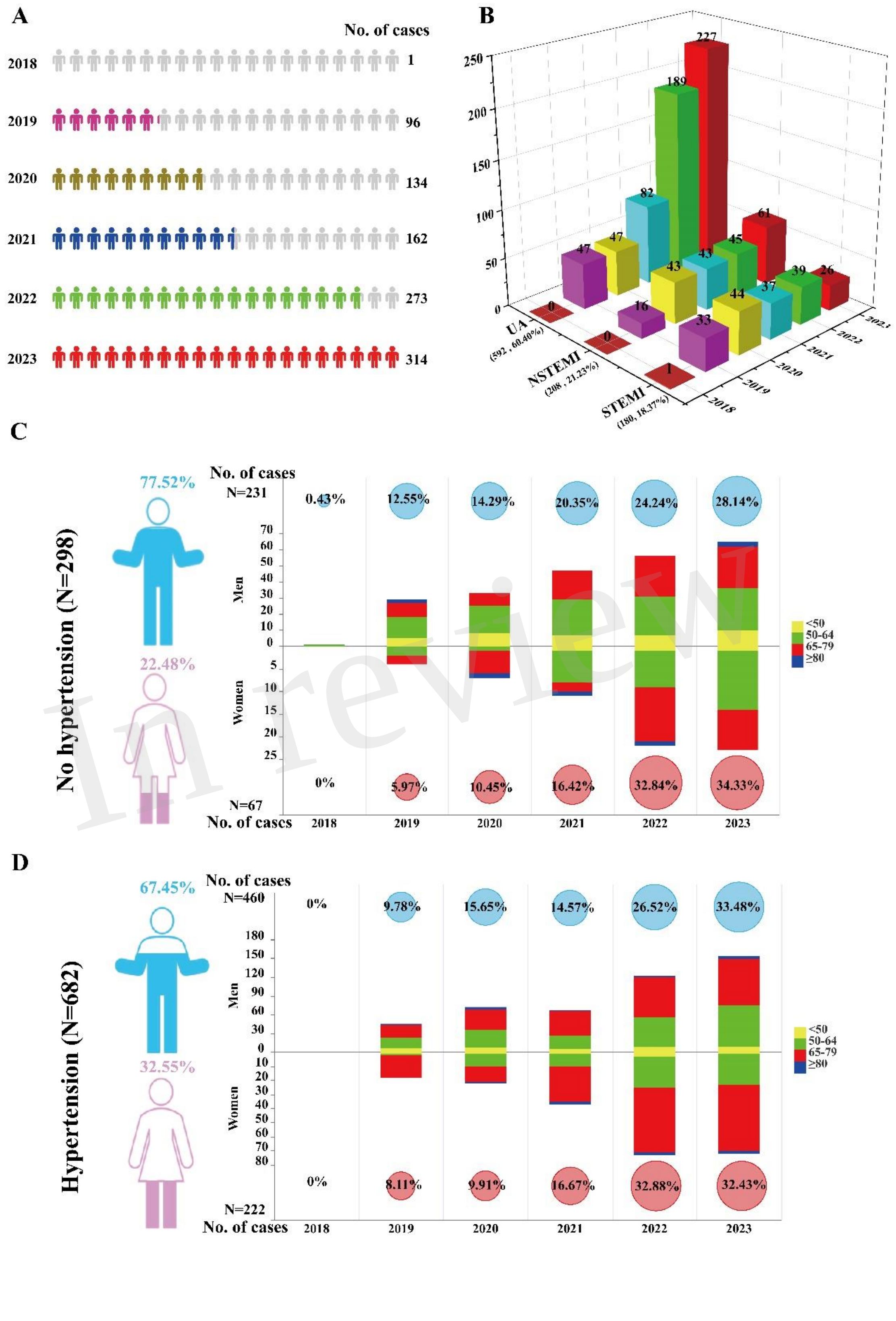

A
No. of cases
2018 1
2019 96
2020 134
2021 162
2022 273
2023 314
B
UA (592, 60.40%)
NSTEMI (208, 21.23%)
STEMI (180, 18.37%)
C
No hypertension (N=298)
77.52%
22.48%
No. of cases
N=231
N=67
Men
Women
<50
50-64
65-79
≥80
D
Hypertension (N=682)
67.45%
32.55%
No. of cases
N=460
N=222
Men
Women
<50
50-64
65-79
≥80

Figure 2

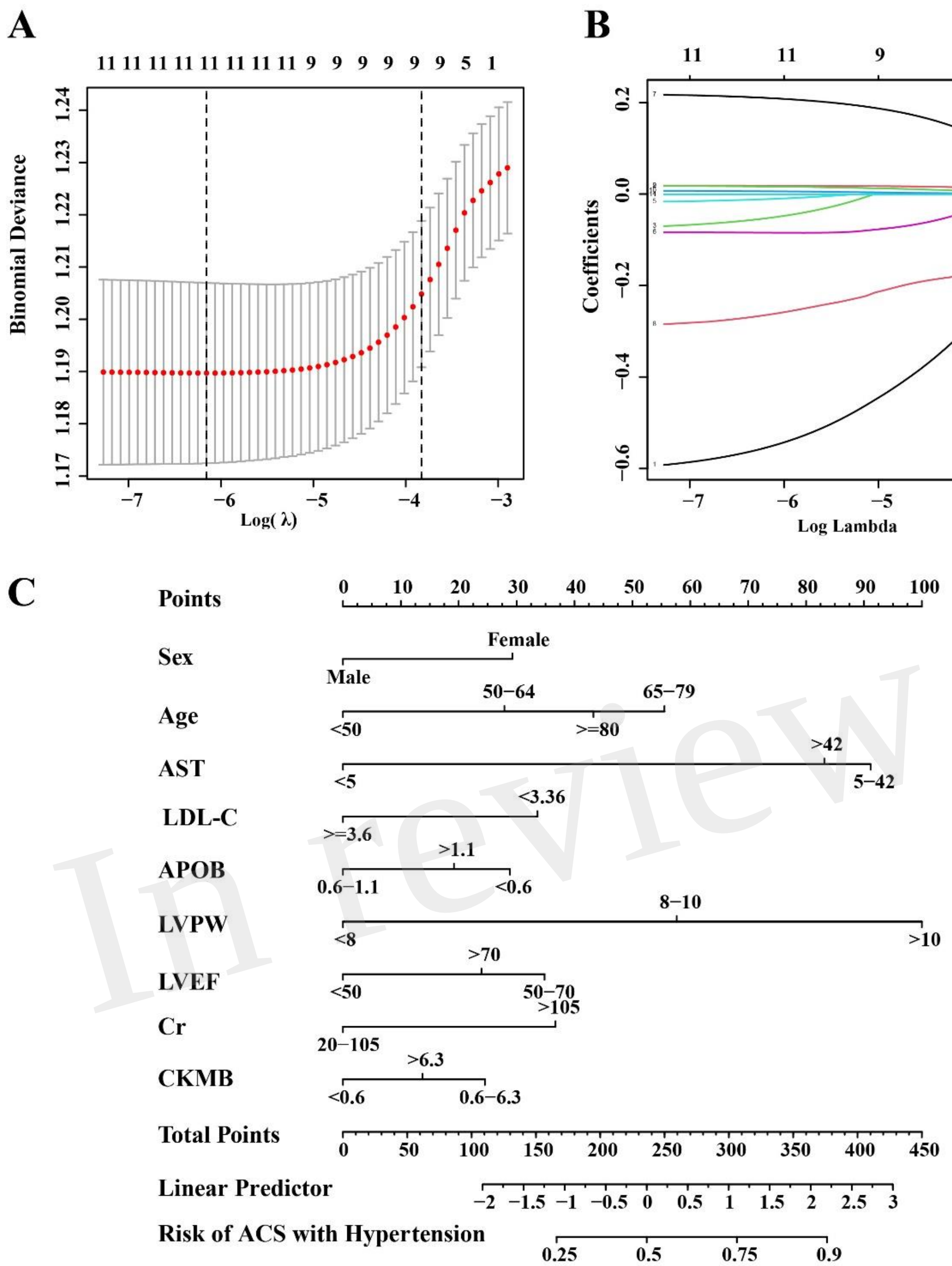
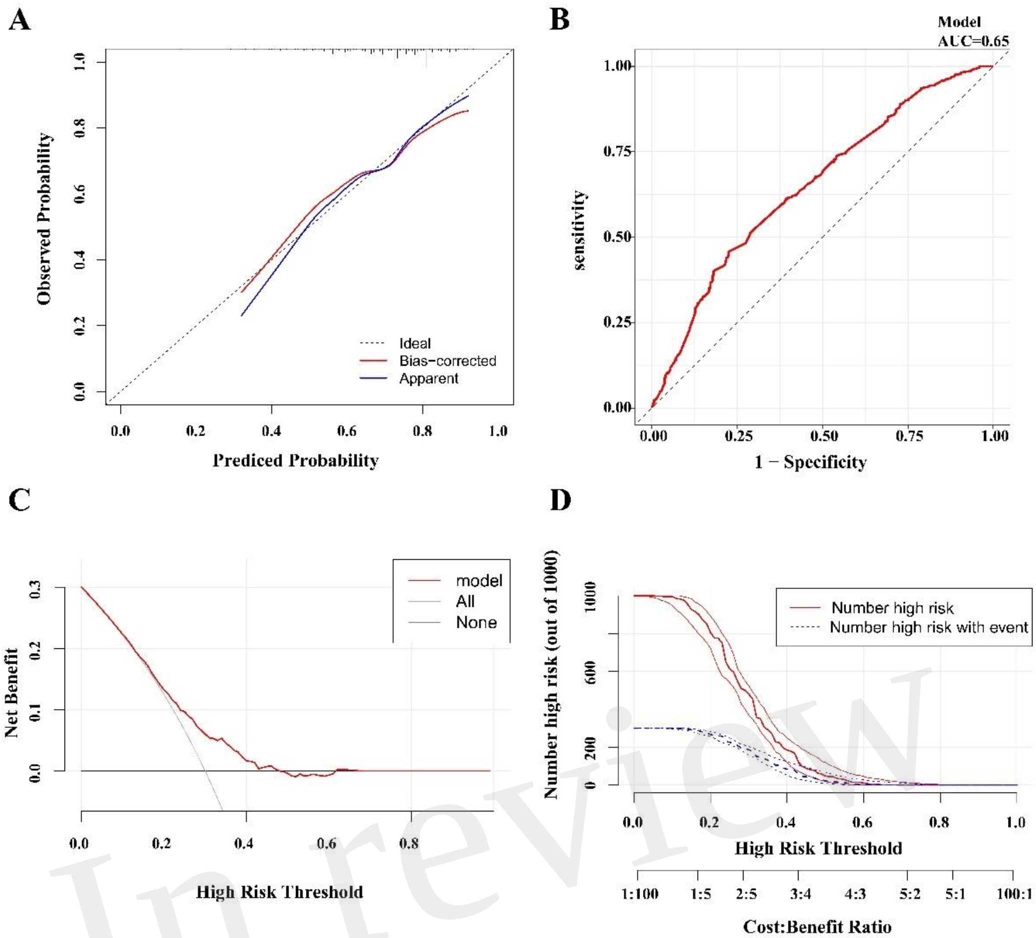

A
11 11 11 11 11 11 11 11 9 9 9 9 9 9 5 1
Binomial Deviance
1.17
1.18
1.19
1.20
1.21
1.22
1.23
1.24
−7
−6
−5
−4
−3
Log(λ)
B
11 11 9 9 1
Coefficients
0.2
0.0
−0.2
−0.4
−0.6
−7
−6
−5
−4
−3
Log Lambda
C
Points
0 10 20 30 40 50 60 70 80 90 100
Sex
Female
Male
Age
50−64
65−79
<50
>=80
AST
>42
<5
5−42
LDL-C
<3.36
>=3.6
APOB
>1.1
0.6−1.1
<0.6
LVPW
8−10
<8
>10
LVEF
>70
<50
50−70
Cr
>105
20−105
CKMB
>6.3
<0.6
0.6−6.3
Total Points
0 50 100 150 200 250 300 350 400 450
Linear Predictor
−2 −1.5 −1 −0.5 0 0.5 1 1.5 2 2.5 3
Risk of ACS with Hypertension
0.25 0.5 0.75 0.9

Figure 3


A
Observed Probability
Prediced Probability
Ideal
Bias−corrected
Apparent
B
Model
AUC=0.65
sensitivity
1 − Specificity
C
Net Benefit
High Risk Threshold
model
All
None
D
Number high risk (out of 1000)
Number high risk
Number high risk with event
High Risk Threshold
1:100
1:5
2:5
3:4
4:3
5:2
5:1
100:1
Cost:Benefit Ratio